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1 Introduction

AgriConnect is a digital marketplace designed to connect local farmers directly with wholesalers, eliminating intermediaries such as commission agents. This platform ensures fair pricing for farmers, provides wholesalers with access to fresh produce, and promotes agricultural sustainability through transparent communication and efficient workflows. The project includes a Flutter-based mobile application for farmers and buyers, a Reactbased admin

dashboard, and a .NET Core backend with PostgreSQL database. The backend is deployed on Oracle Cloud, and the admin website is hosted on Firebase.

This report outlines the project's objectives, functionalities, modules, and deployment details, followed by a comprehensive user manual to guide users in navigating and utilizing AgriConnect's features effectively.

2 Project Overview

AgriConnect aims to revolutionize the agricultural supply chain by providing a B2B (Businessto-Business) platform where farmers can sell their crops directly to wholesalers. The platform supports features like crop listing, order management, real-time payments via Payfast, and AIpowered chatbot assistance using Gemini. It also includes educational resources to assist nontech-savvy users and advanced analytics for platform oversight.

2.1 Objectives

- Connect farmers and wholesalers directly to ensure fair pricing and transparency. • Reduce costs for wholesalers and local shopkeepers by eliminating intermediaries.
- Promote agricultural sustainability through efficient supply chain management.
- Provide user-friendly interfaces and support features like chatbots and educational resources.
- Ensure secure transactions and scalable platform operations.

3 System Architecture

AgriConnect is built using a modern tech stack to ensure scalability, security, and usability:

- **Frontend (Mobile App)**: Flutter for cross-platform mobile application (Android and iOS) for farmers and buyers.
- **Frontend (Admin Dashboard)**: React-based web application hosted on Firebase.
- **Backend**: .NET Core Web API deployed on Oracle Cloud.
- **Database**: PostgreSQL for storing user data, crop listings, and order details.
- **Payment Gateway**: Payfast for secure and real-time online payments.
- **AI Integration**: Gemini-based chatbot for user support and query resolution.

4 Modules and Functionalities

AgriConnect is divided into three primary modules: Farmer, Buyer, and Admin. Each module has specific functionalities tailored to the needs of its users.

4.1 Farmer Module

The Farmer module allows farmers to manage their crop listings, track sales, communicate with buyers, and access support features.

- **Home Screen (Tab 1)**: Displays a list of the farmer's current crop listings, including crop name, description, price, and availability.
- **Add Crops (Tab 2)**: Enables farmers to add new crops with details such as name, description, price, quantity, and photos. Farmers can also edit or delete existing listings.
- **Profile (Tab 3)**: Allows farmers to view and update their profile information, including contact details and account settings.
- **Chatbot (Tab 4)**: Integrates a Gemini-based AI chatbot to assist with croprelated queries, platform navigation, and troubleshooting.
- **Drawer Options**:
  - *Home*: Redirects to the main screen with crop listings.
  - *Profile*: Access to profile management.
  - *Add Crops*: Shortcut to the crop addition screen.
  - *Confirm Order*: Displays pending orders awaiting confirmation.
  - *Completed Order*: Shows a history of completed orders.
  - *Messages*: Enables direct messaging with buyers for order discussions.
  - *Chatbot*: Access to the Gemini-based chatbot for support.

4.2 Buyer Module

The Buyer module enables wholesalers to browse crops, manage their cart, and complete purchases.

- **All Crops (Tab 1)**: Displays a catalog of available crops with search and filter options by category or keyword.
- **Profile (Tab 2)**: Allows buyers to view and update their profile information.
- **Cart (Tab 3)**: Shows items added to the cart with options to adjust quantities or remove items. Includes a checkout button for payment via Payfast.
- **Chatbot (Tab 4)**: Provides access to the Gemini-based chatbot for price suggestions, quality inquiries, and platform guidance.
- **Drawer Options**:
  - *Home*: Redirects to the crop catalog.
  - *Profile*: Access to profile management.
  - *Messaging*: Enables direct communication with farmers.

- *Completed Order*: Displays a history of completed orders.
- *Confirmed Order*: Shows orders confirmed by farmers.
- *Transactions*: Lists payment history and transaction details.
- *Chatbot*: Access to the Gemini-based chatbot.

4.3 Admin Module

The Admin module, accessible via a React-based web dashboard, allows platform administrators to oversee operations and manage users.

- **User Management**: View and manage farmer and buyer accounts, including registration approvals and account suspensions.
- **Order Management**: Monitor the status of all orders (Not Confirmed, Confirmed, Completed).
- **Platform Activity**: Analyze sales data, crop availability, and user activity through advanced analytics.
- **Instructor Assignment**: Assign instructors to train farmers facing difficulties using the platform.
- **Feedback Review**: Review user complaints and feedback to ensure platform integrity.

4.4 Additional Features

- **Payfast Integration**: Secure and real-time payment processing for buyers during checkout.
- **Gemini-based Chatbot**: Provides automated support for farmers and buyers, assisting with crop queries, pricing suggestions, and platform navigation.
- **Educational Resources**: Videos and guides to help farmers learn how to use the platform effectively.
- **Order Workflow**: Buyers add crops to the cart, proceed to checkout, and pay via Payfast. Orders move to “Confirmed” upon farmer approval and “Completed” upon delivery confirmation.

5 Business Model

AgriConnect operates on a B2B model, facilitating direct transactions between farmers and wholesalers. Additional revenue streams include:

- **Subscription-Based Services**: Premium features for farmers, such as advanced analytics and highlighted listings.
- **Service Fee**: A small commission on each transaction to sustain platform operations.

6 Novelty

AgriConnect addresses unique challenges in the agricultural sector, particularly in regions like Pakistan, by:

- Eliminating intermediaries to ensure fair pricing.
- Providing a platform tailored to non-tech-savvy users with simplified workflows and educational resources.
- Integrating AI-driven chatbot support for real-time assistance.
- Promoting transparency and sustainability in the agricultural supply chain.

7 Deployment Details

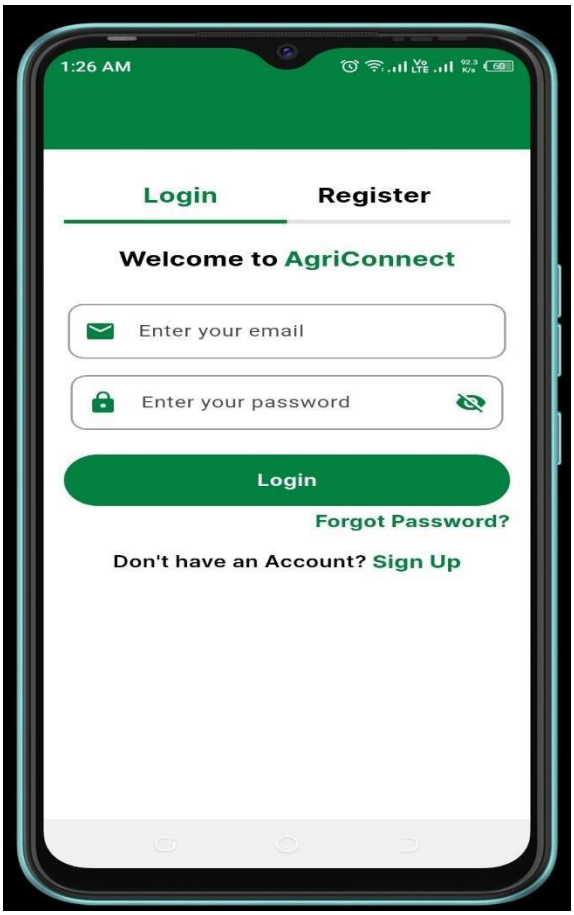
- **Mobile App**: Deployed using Flutter, compatible with Android and iOS.
- **Admin Dashboard**: Hosted on Firebase using React.
- **Backend**: Deployed on Oracle Cloud using .NET Core Web API.
- **Database**: PostgreSQL for data storage.
- **Payment Gateway**: Payfast for secure transactions.

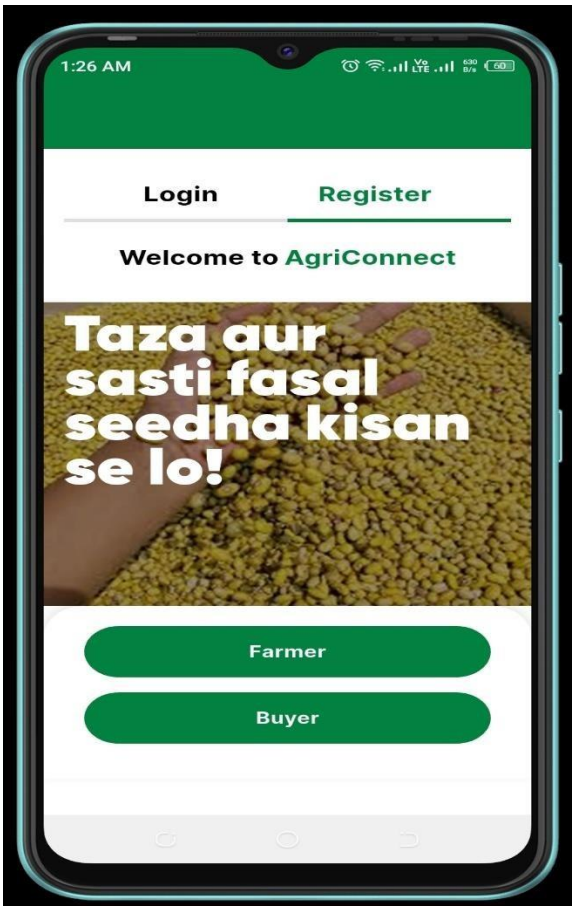
8 User Manual

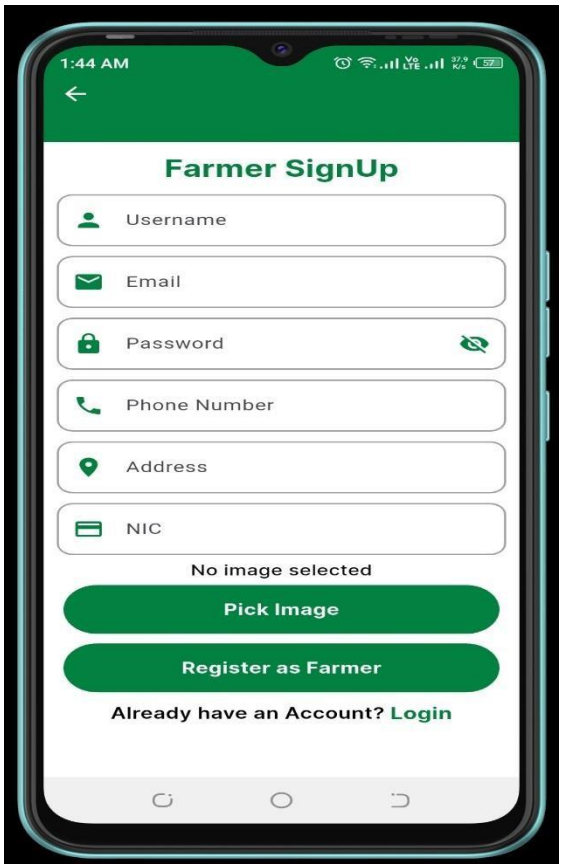
This user manual provides step-by-step instructions for farmers, buyers, and admins to use AgriConnect effectively.

8.1 Farmer User Manual

1. **Register/Login**:
    - Open the AgriConnect mobile app.
    - Click “Register” to create an account or “Login” with existing credentials.
    - Provide required details (name, contact, address) during registration.

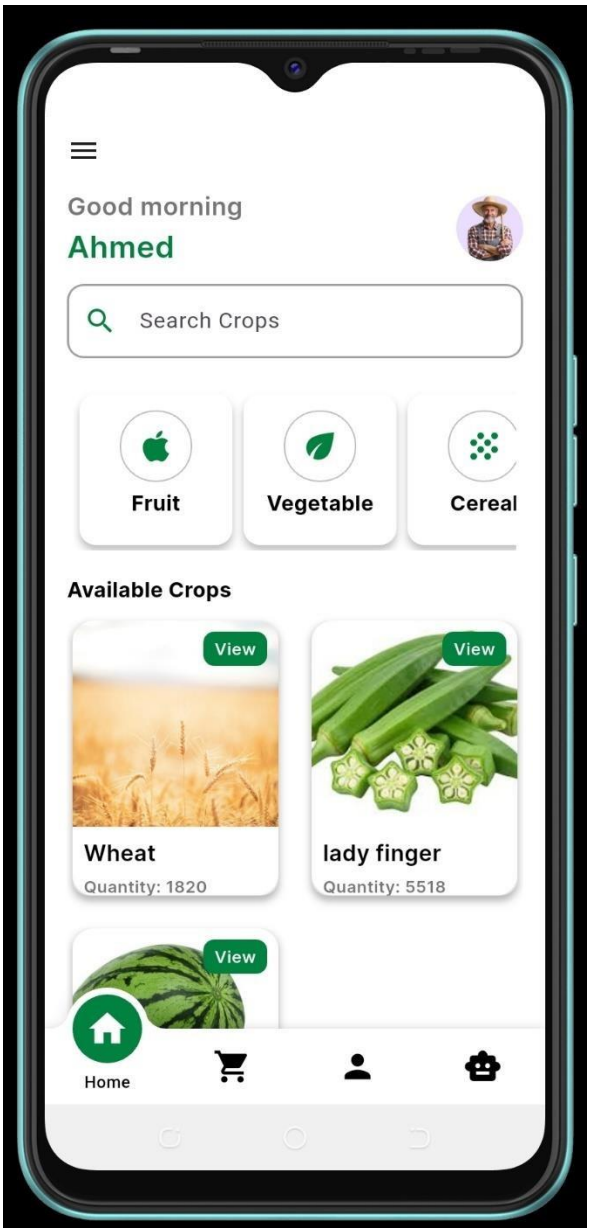




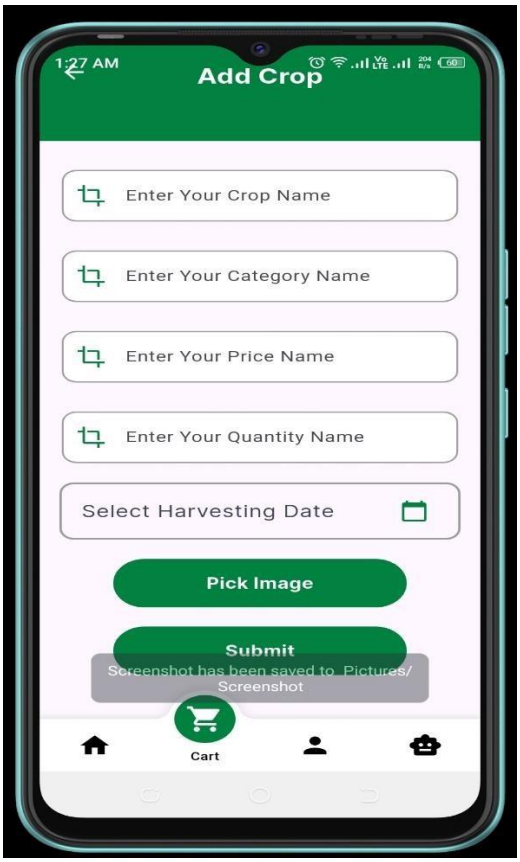


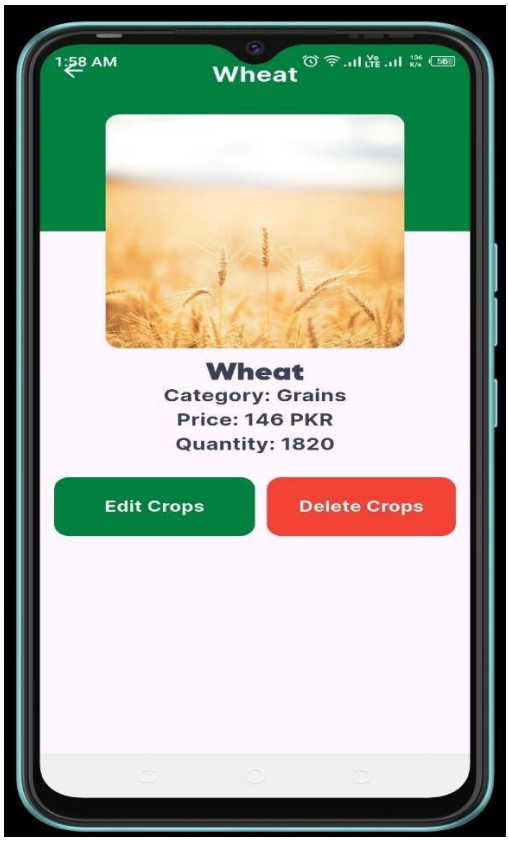
2. **View Crop Listings (Home Screen)**:

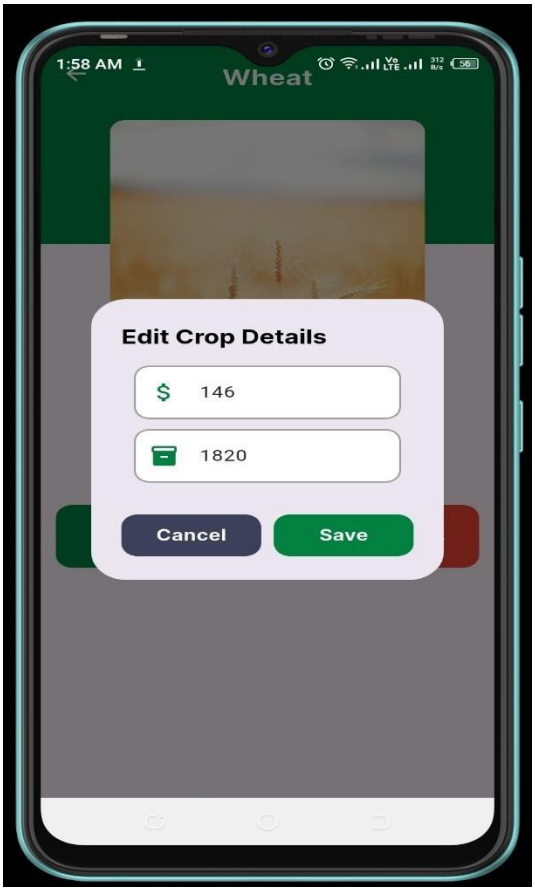
- On the first tab, view all your listed crops with details like name, price, and quantity.



3. **Add/Edit/Delete Crops**:
    - Navigate to the second tab (“Add Crops”).
    - Click “Add Crop” to enter details (name, description, price, quantity, photos).
    - To edit/delete, select a crop from the list and choose the appropriate action.

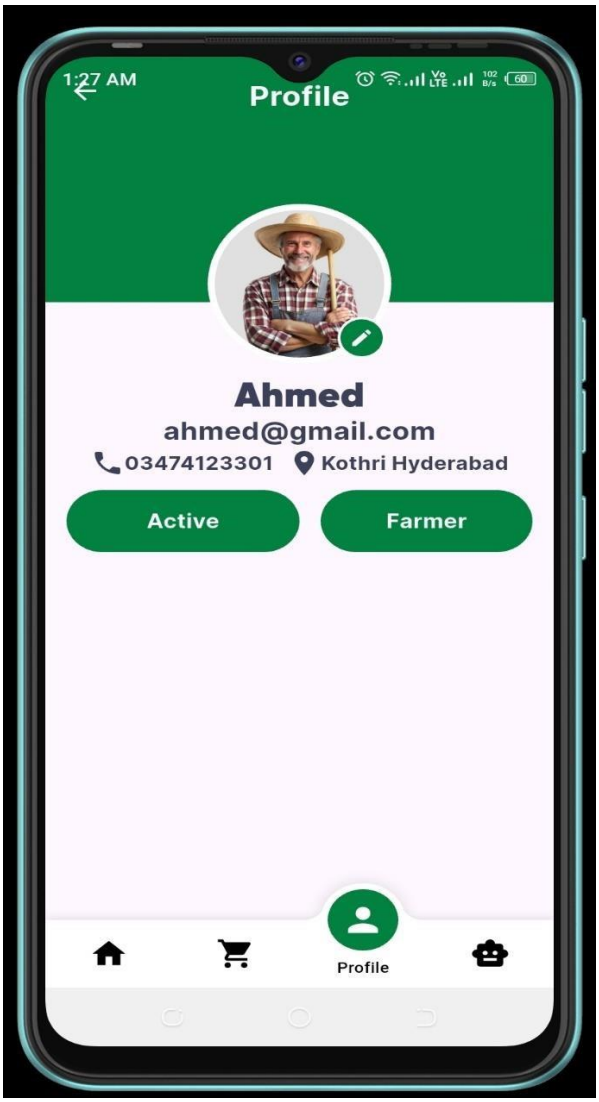






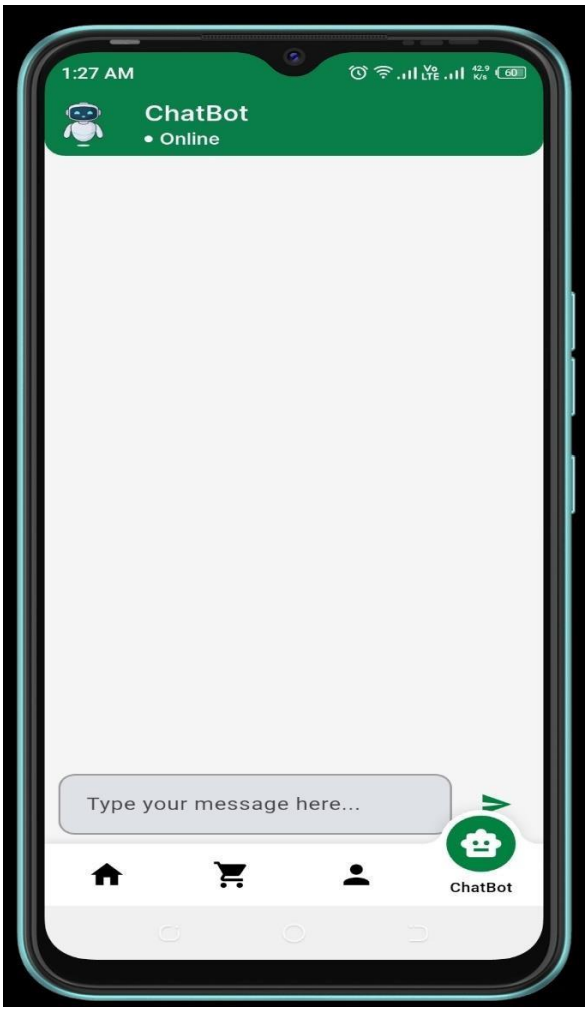
4. **Manage Profile**:

• Go to the third tab (“Profile”) to view or update your account details.

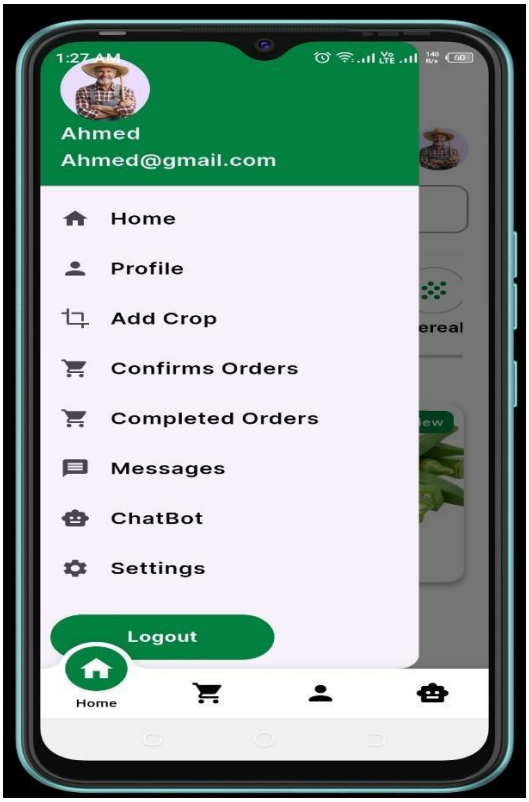


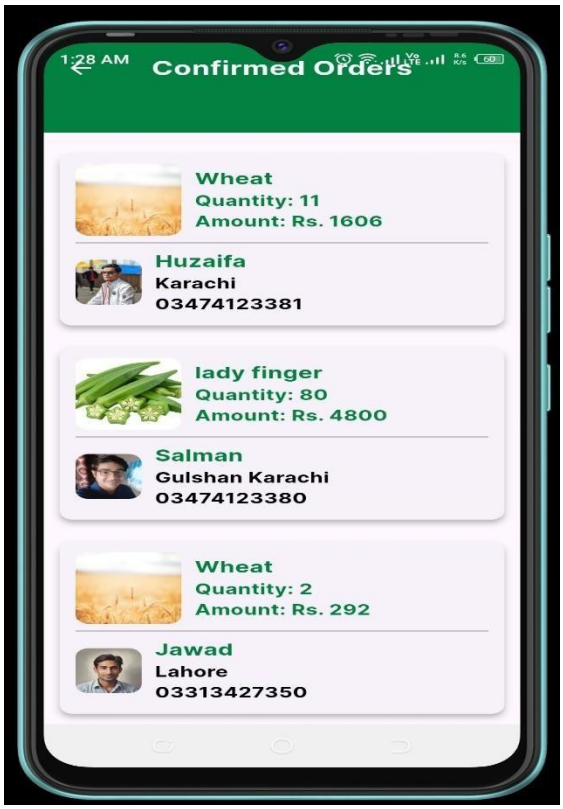
5. **Use Chatbot**:

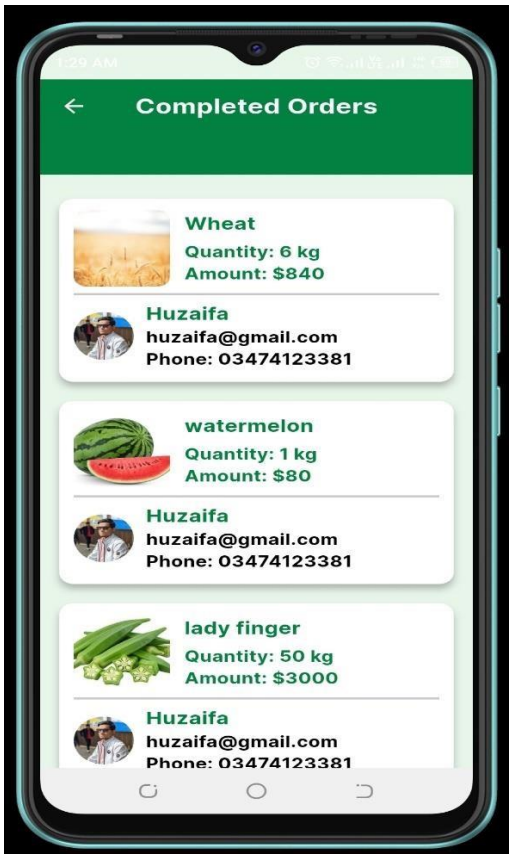
   - Access the fourth tab (“Chatbot”) to ask questions about crops, pricing, or platform usage.
   - The Gemini-based chatbot will provide real-time responses.



6. **Drawer Navigation**:
    - Open the drawer to access:
        - *Confirm Order*: View and confirm pending orders from buyers.
        - *Completed Order*: Check the history of completed orders.
        - *Messages*: Chat with buyers to discuss orders.
        - *Chatbot*: Access the chatbot for additional support.

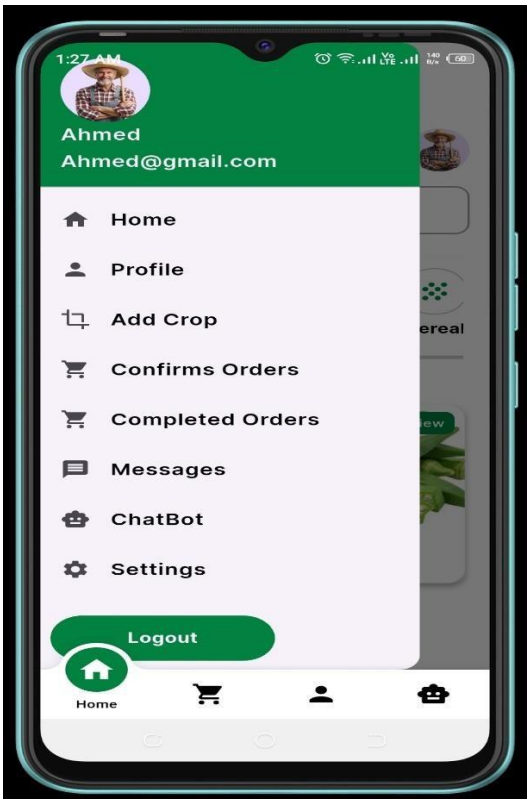

1:27 AM
Ahmed
Ahmed@gmail.com
Home
Profile
Add Crop
Confirms Orders
Completed Orders
Messages
ChatBot
Settings
Logout
Home


1:28 AM
Confirmed Orders
Wheat
Quantity: 11
Amount: Rs. 1606
Huzaifa
Karachi
03474123381
lady finger
Quantity: 80
Amount: Rs. 4800
Salman
Gulshan Karachi
03474123380
Wheat
Quantity: 2
Amount: Rs. 292
Jawad
Lahore
03313427350


Completed Orders
Wheat
Quantity: 6 kg
Amount: $840
Huzaifa
huzaifa@gmail.com
Phone: 03474123381
watermelon
Quantity: 1 kg
Amount: $80
Huzaifa
huzaifa@gmail.com
Phone: 03474123381
lady finger
Quantity: 50 kg
Amount: $3000
Huzaifa
huzaifa@gmail.com
Phone: 03474123381

7. **Logout**:

    • Use the drawer to log out of the app.



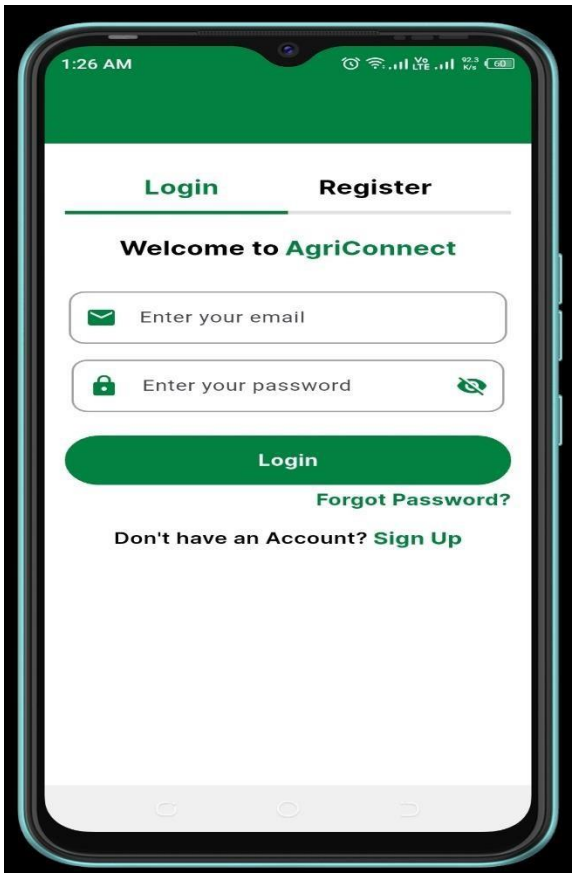
8.2 Buyer User Manual

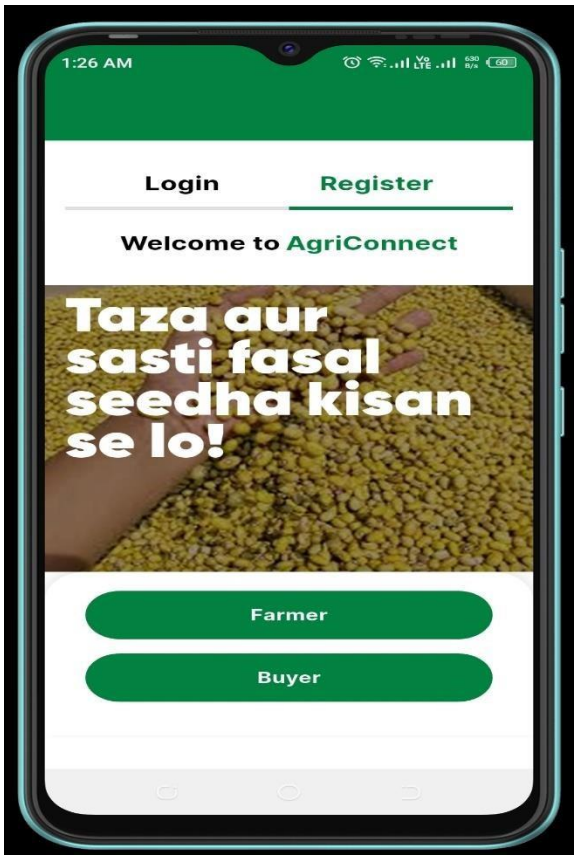
1. **Register/Login**:

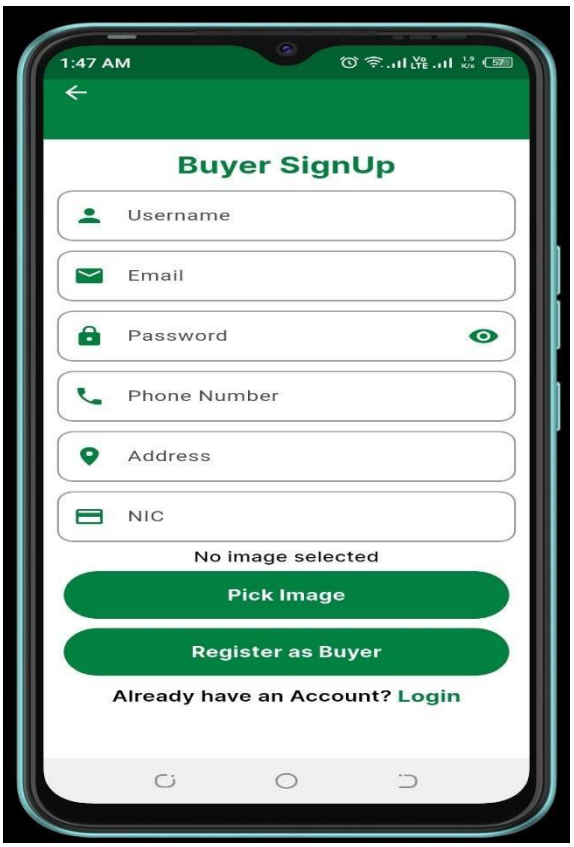
    • Open the AgriConnect mobile app.

    • Click “Register” to create an account or “Login” with existing credentials.

    • Provide required details (name, contact, address) during registration.

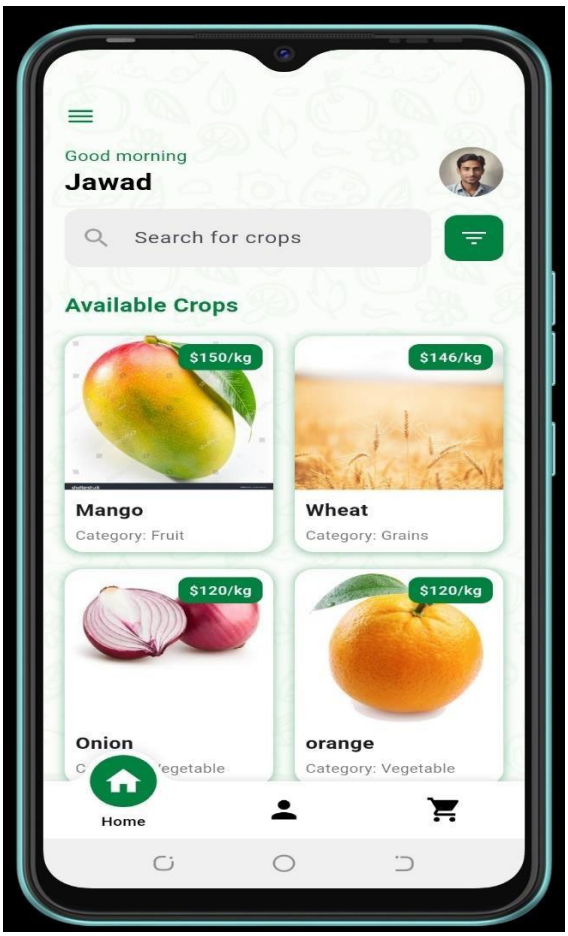






2. **Browse Crops**:

- On the first tab (“All Crops”), browse or search for crops by category or keyword.
- Click on a crop to view details (description, price, farmer contact).



3. **Add to Cart**:
   - Select a crop, choose the quantity, and click “Add to Cart”.
   - Navigate to the third tab (“Cart”) to review selected items.

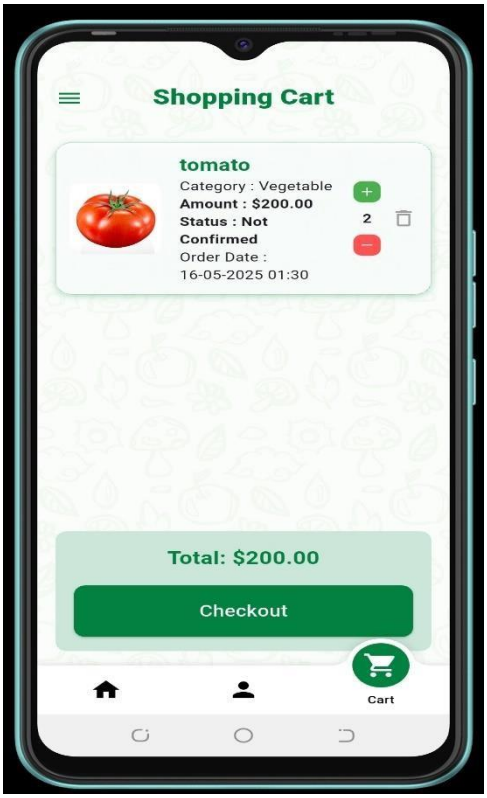


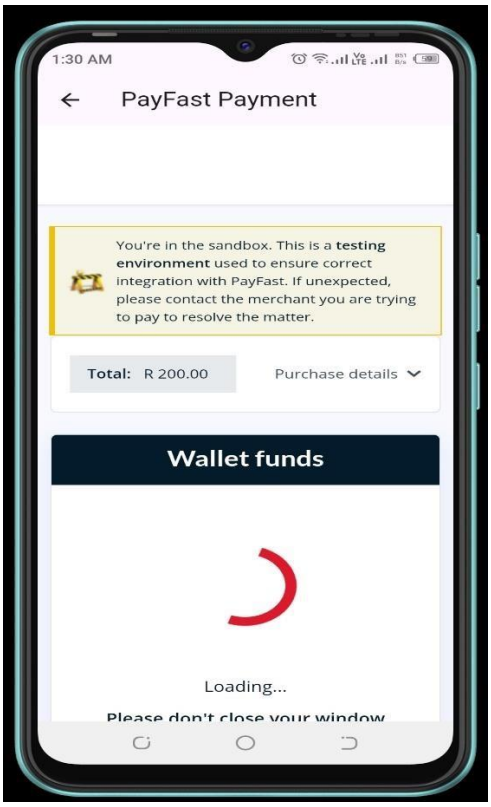
4. **Checkout and Payment**:

- In the cart, click “Checkout” to proceed to payment.

  Complete the payment via Payfast.

- Once paid, the order moves to “Confirmed Orders” upon farmer approval.





5. **Use Chatbot**:

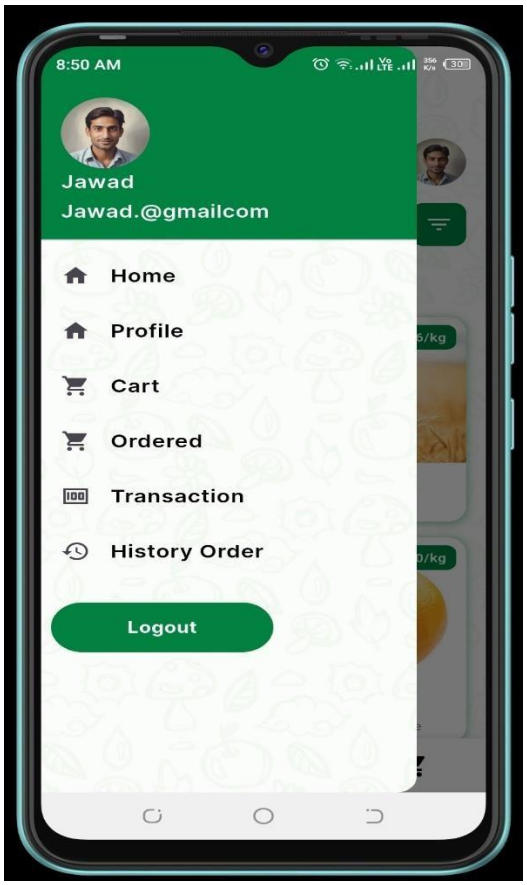
   • Access the fourth tab (“Chatbot”) for price suggestions or quality inquiries.

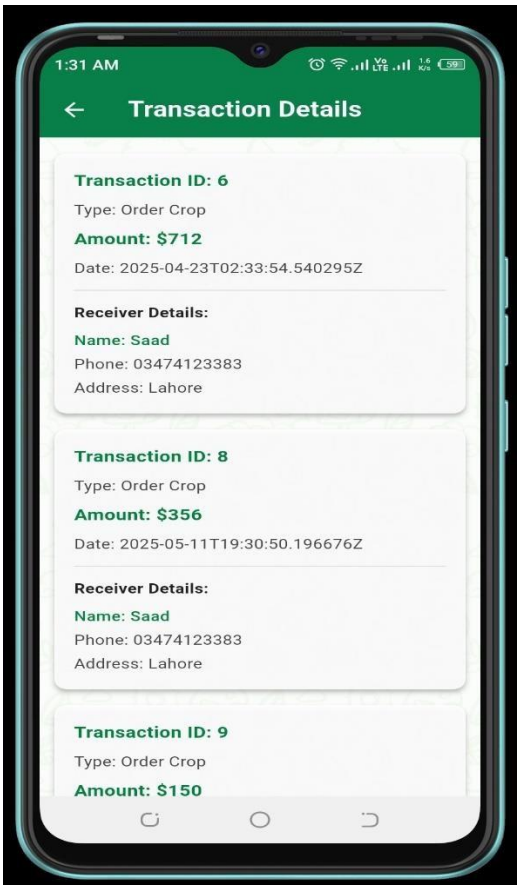


6. **Drawer Navigation**:

• Open the drawer to access:

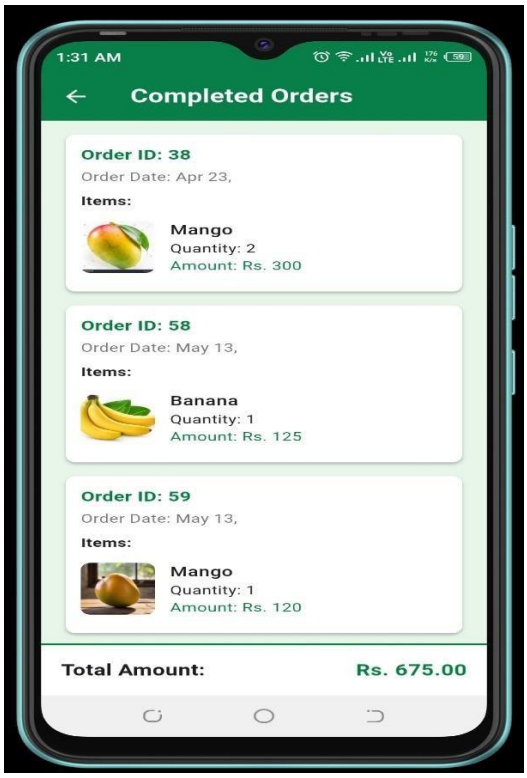
  – *Messaging*: Communicate with farmers about orders.
  – *Confirmed Order*: View orders confirmed by farmers.
  – *Completed Order*: Check completed order history.
  – *Transactions*: Review payment history.
  – *Chatbot*: Access the chatbot for support.





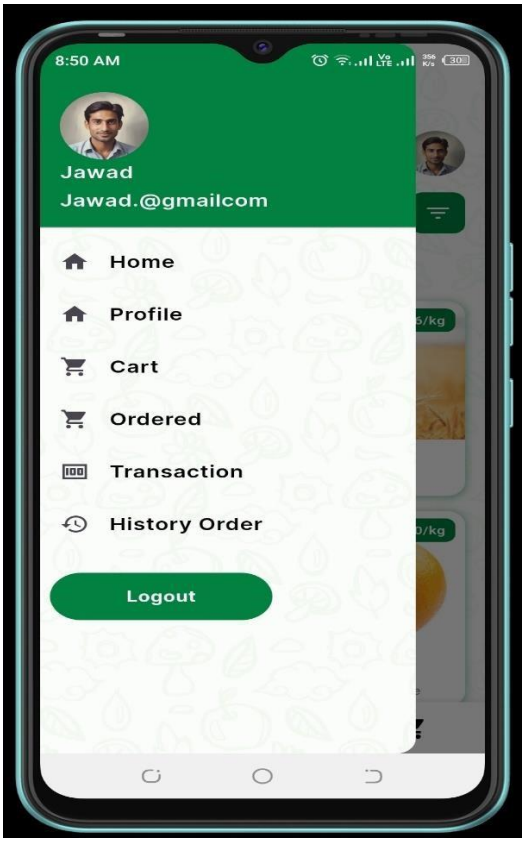
7. **Complete Order**:

   • In “Confirmed Orders,” mark an order as “Completed” upon receiving the delivery.



8. **Logout**:

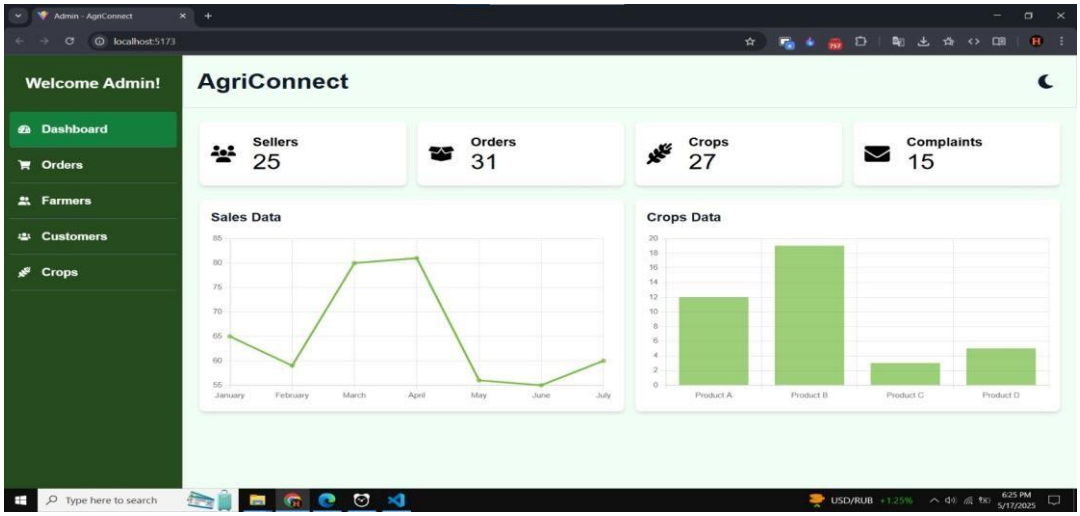
    • Use the drawer to log out of the app.



8.3 Admin User Manual

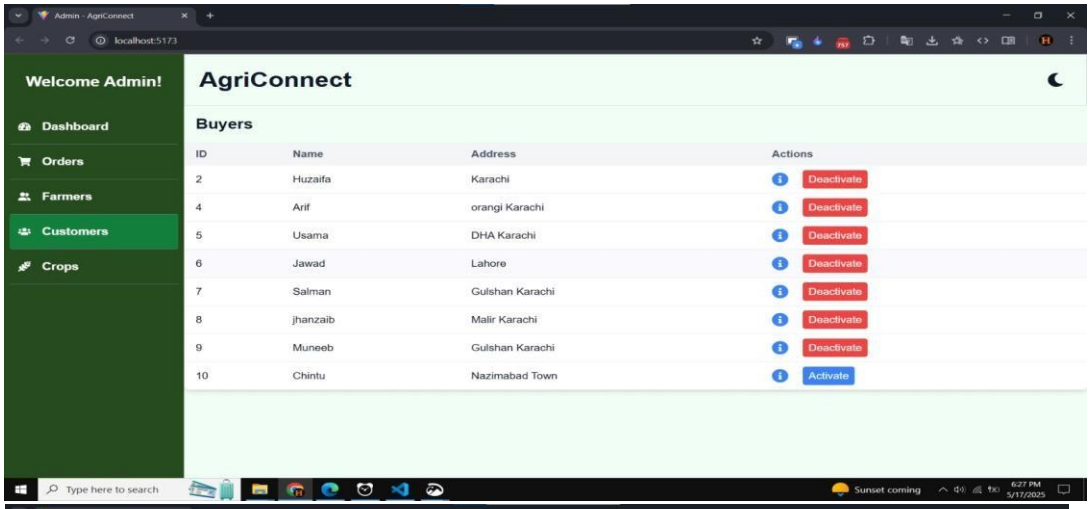
1. **Access Admin Dashboard**:

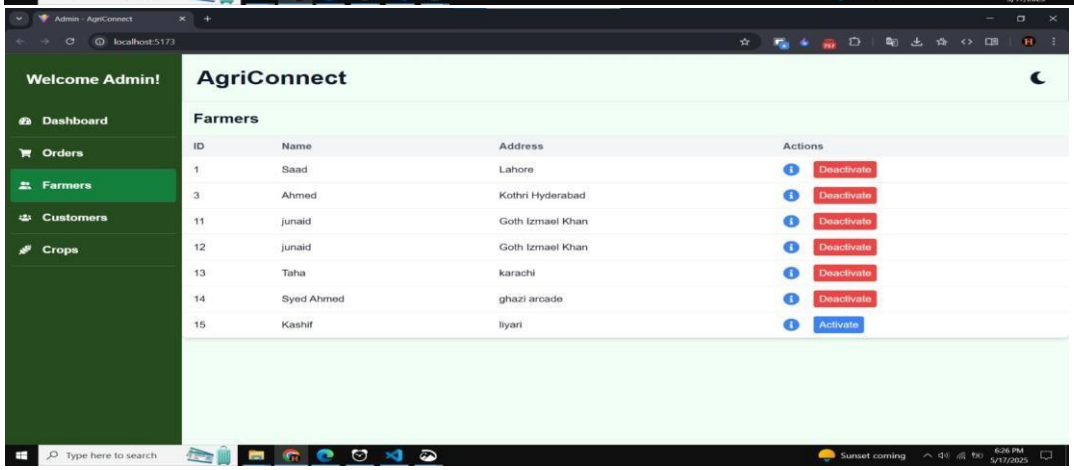
- Open the browser and navigate to the AgriConnect admin dashboard URL (hosted on Firebase).
- View total sellers, orders, crops, and complaints, along with monthly sales and crop distribution charts.



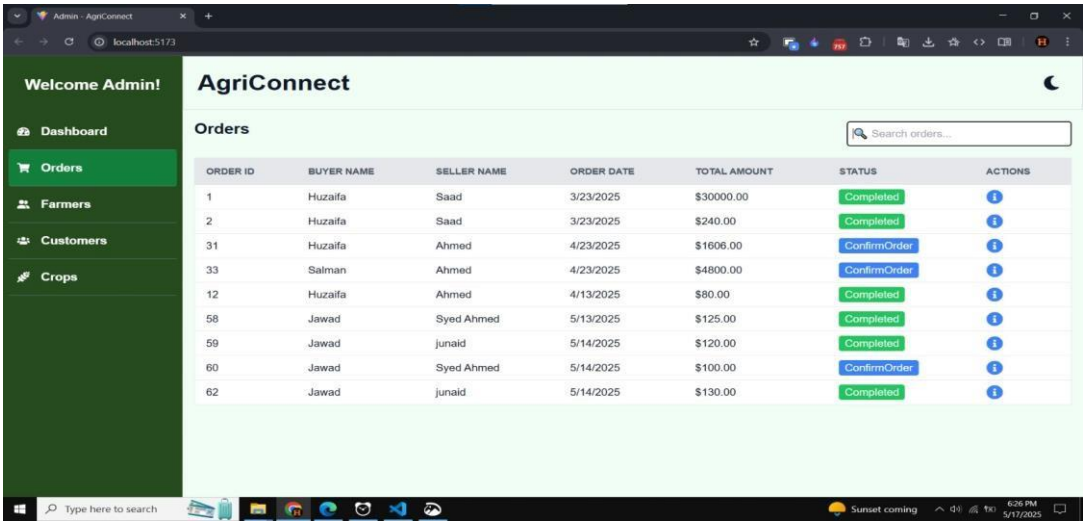
2. **Manage Users**:

- View lists of registered farmers and buyers.
- Approve, suspend, or delete user accounts as needed.

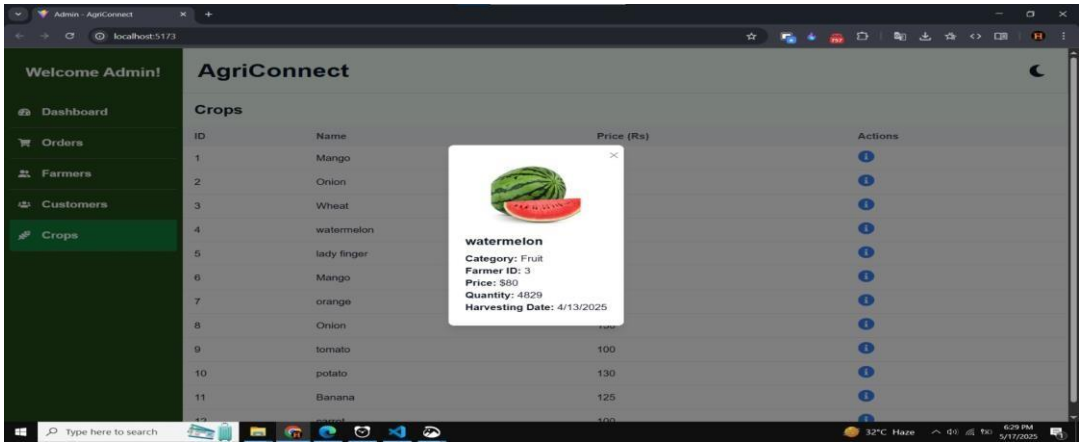




3. **Monitor Orders**:
    - Check the status of all orders (Not Confirmed, Confirmed, Completed).
    - View order details, including farmer and buyer information.



4. **Crop Details View**:
    - Display complete crop details like name, category, farmer ID, price, quantity, and harvesting date in a pop-up view.



9 Installation Instructions

For your teacher or any user to set up and test AgriConnect locally, follow these steps:

9.1 Flutter Mobile App

1. **Requirements**:
    - Flutter SDK (version 3.x)
    - Dart SDK
    - Android Studio or Visual Studio Code with Flutter plugin
    - Android Emulator or physical Android device
    - Chrome (for Flutter web testing)

2. **Steps**:
    (a) Clone the repository: git clone https://github.com/MuhammadSaadEhsan/AgriConnect
    (b) Navigate to the project folder: cd AgriConnect
    (c) Install dependencies: flutter pub get
    (d) Run the app: flutter run

9.2 Admin Dashboard (React)

1. **Requirements**:
    - Node.js (v16 or later)
    - npm or yarn
2. **Steps**:
    (a) Clone the repository: git clone https://github.com/huzaifaalisiddiqui/admin_dashb
    (b) Navigate to the project folder: cd admin_dashboard_agriconnect
    (c) Install dependencies: npm install
    (d) Start the app: npm start

9.3 Backend (.NET Core)

1. **Requirements**:
    - .NET SDK (6.0 or later)

      Visual Studio 2022 or Visual Studio Code
    - PostgreSQL
    - Postman (optional, for API testing)
2. **Steps**:
    (a) Clone the backend repository: git clone <BACKEND_REPO_URL>
    (b) Navigate to the project folder: cd <project-folder>
    (c) Restore NuGet packages: dotnet restore
    (d) Update the database: dotnet ef database update
    (e) Run the application: dotnet run
    (f) Update appsettings.json with the PostgreSQL connection string.
    (g) Ensure CORS is enabled for frontend requests.
    (h) Access Swagger UI at https://localhost:5001/swagger for API documentation.

10 Significance

- **Economic Impact**: Increases farmer profit margins and reduces costs for wholesalers and consumers.
- **Social Impact**: Enhances food accessibility and promotes sustainable agricultural practices.
- **Technological Impact**: Introduces digital solutions to the agricultural sector with scalable and user-friendly features.

11 Conclusion

AgriConnect is a transformative platform that bridges the gap between farmers and wholesalers, fostering a sustainable and transparent agricultural ecosystem. With its intuitive mobile app, robust admin dashboard, and AI-driven support, it addresses the unique needs of the agricultural sector while ensuring accessibility for all users. This report and user manual provide comprehensive guidance for your teacher to explore and evaluate the platform's functionalities.